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MATHS MASTER _____

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CANDIDATE NUMBER

2026 Trial Examination

Form VI Mathematics Extension 1

Wednesday 19th August, 2026

12:50pm

General Instructions

- Reading time — 10 minutes
- Working time — 2 hours
- Attempt all questions.
- Write using black pen.
- Calculators approved by NESA may be used.
- A loose reference sheet is provided separate to this paper.

Fourteen Questions — 70 Marks

Section I (10 marks) Questions 1 – 10

- This section is multiple choice. Each question is worth 1 mark.
- Record your answers on the provided answer sheet.

Section II (60 marks) Questions 11 – 14

- Each question is worth 15 marks.
- Relevant mathematical reasoning and calculations are required.
- Start each question in a new booklet.

Collection

- Your name and master should only be written on this page.
- Write your candidate number on this page, on each booklet and on the multiple-choice answer sheet.
- If you use multiple booklets for a question, place them inside the first booklet for the question.
- Arrange your solutions in order.

Checklist

- Reference sheet
- Multiple-choice answer sheet
- 4 booklets per boy
- Candidature: 131 pupils

Writer: OMD

Section I

1. Let $\underline{a} = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$. What is the value of $|\underline{a}|$? 1
- (A) $\sqrt{5}$
- (B) 5
- (C) $\sqrt{13}$
- (D) 13
2. If $f(x) = \sin^{-1}(x - 1)$, which of the following is the range of $f(x)$? 1
- (A) $[-\pi, \pi]$
- (B) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
- (C) $[-1, 1]$
- (D) $[0, 2]$
3. Which of the following vectors is perpendicular to $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$? 1
- (A) $\begin{pmatrix} -2 \\ 3 \end{pmatrix}$
- (B) $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$
- (C) $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$
- (D) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$
4. A headmaster realises that there is at least one month of the year when at least 34 pupils have a birthday. 1
- What is the minimum possible number of pupils at the school?
- (A) 396
- (B) 397
- (C) 408
- (D) 409

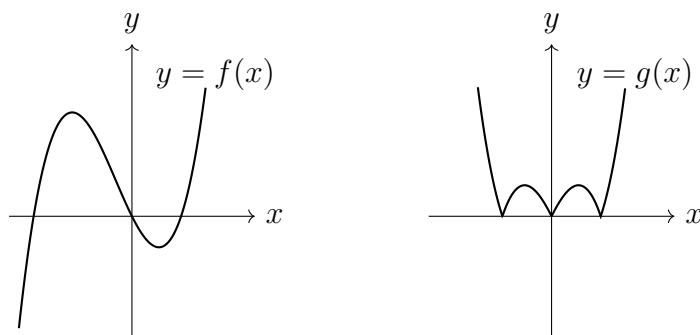
5. A particle moves so that $x = \tan t$ and $y = x^2$. Which of the following is equal to $\frac{dy}{dt}$? 1

- (A) $2 \sec^2 t$
 (B) $2 \tan t$
 (C) $2 \tan t \sec^2 t$
 (D) $\sec^2 t$

6. Let $\underline{a} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. The projection of $\underline{a}$ onto $\underline{b}$ is $\lambda \underline{b}$. What is the value of λ ? 1

- (A) $\frac{1}{5}$
 (B) $\frac{2}{5}$
 (C) $\frac{4}{5}$
 (D) 1

7. The graphs of $y = f(x)$ and $y = g(x)$ are shown below. 1



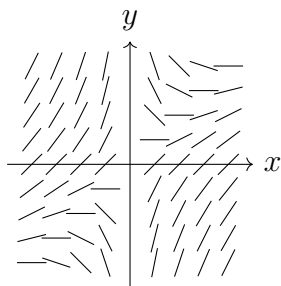
Which transformation maps $f(x)$ to $g(x)$?

- (A) $g(x) = |f(x)|$
 (B) $g(x) = f(|x|)$
 (C) $|g(x)| = f(x)$
 (D) $g(x) = |f(|x|)|$

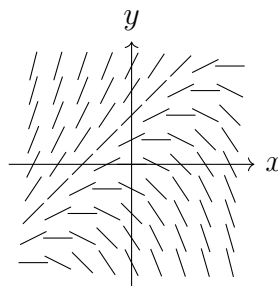
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8. Which of the following slope fields corresponds to the differential equation $\frac{dy}{dx} = \frac{y-x}{x}$? 1

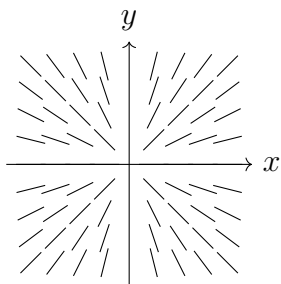
(A)



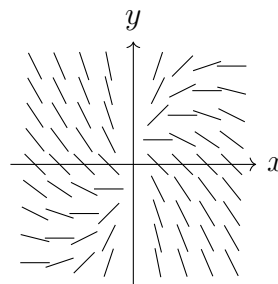
(B)



(C)



(D)



9. The nine members of the Fellowship of the Ring are seated randomly around a circular table. 1

What is the probability that the two youngest members of the Fellowship are seated next to each other?

(A) $\frac{7!}{9!2!}$

(B) $\frac{7!2!}{9!}$

(C) $\frac{7!}{8!2!}$

(D) $\frac{7!2!}{8!}$

10. Let $g(x) = \sin^2 x - \frac{x}{3}$. How many times does the graph of $y = g(x)$ intersect the x -axis? 1

(A) once

(B) twice

(C) three times

(D) five times

End of Section I

The paper continues in the next section

Section II

This section consists of long-answer questions.

Marks may be awarded for reasoning and calculations.

Marks may be lost for poor setting out or poor logic.

Start each question in a new booklet.

QUESTION ELEVEN (15 marks) Start a new answer booklet.

(a) How many four-digit numbers can be formed using the digits 1, 2, 3, 4, 5, 6, 7 without repetition? 1

(b) Evaluate $\int_0^1 \frac{2}{1+x^2} dx$. 2

(c) Use the substitution $t = \tan \frac{\theta}{2}$ to solve $2 \sin \theta + \cos \theta = 2$, for $0 \leq \theta \leq 2\pi$. 2

(d) The curve C is defined parametrically by $x = t^2 + 1$ and $y = t^3 - 3t$. Find the gradient of the tangent to C when $t = 2$. 1

(e) The area, A , of a square is increasing at a rate of $12 \text{ cm}^2 \text{ s}^{-1}$. Find the rate at which the side length, x , is increasing when the side length is 3 cm. 2

(f) (i) Express $\cos 2x + \sin 2x$ in the form $R \cos(2x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. 2

(ii) Hence solve $\cos 2x + \sin 2x = 1$ for $0 \leq x \leq 2\pi$. 2

(g) Let $u = \sqrt{x}$. Hence evaluate $\int_1^4 \frac{1}{\sqrt{x}(1+\sqrt{x})} dx$. 3

The paper continues on the next page

QUESTION TWELVE (15 marks) Start a new answer booklet.

- (a) Solve $\frac{x}{x+1} > 3$. 2
- (b) Let $\underline{a} = \begin{pmatrix} -2 \\ 5 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$.
- (i) Find the angle between $\underline{a}$ and $\underline{b}$. Give your answer in exact form. 2
- (ii) Find λ , given that $\underline{c} = \underline{b} + \lambda \underline{a}$ is perpendicular to $\underline{a}$. 1
- (iii) Find the area of the parallelogram with adjacent sides $\underline{a}$ and $\underline{b}$. 1
- (c) Prove that $\tan \theta = \frac{\sin 2\theta}{1 + \cos 2\theta}$. 2
- (d) Use mathematical induction to prove that $7^n - 1$ is divisible by 6 for all positive integers n . 3
- (e) Let O , A and B be three points. Let $\underline{a} = \overrightarrow{OA}$ and $\underline{b} = \overrightarrow{OB}$. Points P and Q lie on AB such that $AP : PB = 1 : 2$, and $AQ : QB = 2 : 1$.
- (i) Express $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ in terms of $\underline{a}$ and $\underline{b}$. 1
- (ii) Show that $\overrightarrow{PQ} = \frac{1}{3}(\underline{b} - \underline{a})$. 1
- (iii) Let M be the midpoint of AB . Use vectors to show that $PM = MQ$. 2

QUESTION THIRTEEN (15 marks) Start a new answer booklet.

- (a) The coefficients of x^2 and x^5 are equal in the expansion of $(x + a)^8$. Find the value of the positive constant a . **2**
- (b) (i) Find $\frac{d}{dx}(x \tan^{-1}(3x))$. **1**
- (ii) Hence find $\int \tan^{-1}(3x) dx$. **2**
- (c) The population, P , of porgs on the remote island of Ahch-To satisfies $\frac{dP}{dt} = \frac{1}{2}P \left(1 - \frac{P}{2000}\right)$, where t is the time in years. Initially there are 300 porgs.
- (i) Find an expression for P in terms of t . You may assume that **3**
- $$\frac{2000}{P(2000 - P)} = \left(\frac{1}{P} + \frac{1}{2000 - P}\right).$$
- (ii) Find the greatest value of $\frac{dP}{dt}$. You do **not** need to justify that it is a maximum. **1**
- (d) (i) Sketch the curves $y = \cos x$ and $y = \sin x$ for $0 \leq x \leq \pi$. Shade the region bounded by $y = \cos x$, $y = \sin x$, and the y -axis. **2**
- (ii) The shaded region from part (i) is rotated about the x -axis. Find the volume of the resulting solid. **4**

The paper continues on the next page

QUESTION FOURTEEN (15 marks) Start a new answer booklet.

(a) In how many arrangements of the letters of the word TATOOINE are the Ts separated by one or more letters? 2

(b) The graph, $y = P(x)$, of the quartic polynomial $P(x) = x^4 - 10x^3 + bx^2 + cx + 16$, touches, but does not cross, the x -axis at two distinct points.

(i) Explain why $P(x) = (x - m)^2(x - n)^2$ for some real numbers m and n . 1

(ii) Given that $mn < 0$, find the possible values of m and n . 3

(c) Use the substitution $y = u(x) \cos x$, where u is some function of x , to solve the differential equation 3

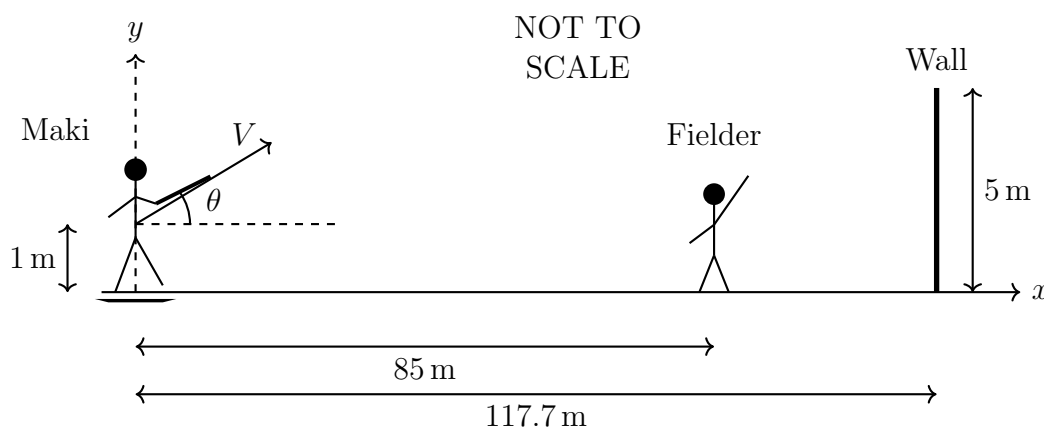
$$y' \cos x = y \sin x + \cos x \sin x.$$

The question continues on the next page

- (d) Shugo Maki is a baseball player. He hits a ball from a point 1 m above horizontal ground. The point at which the ball is hit is referred to as the point of projection. Take the origin, O , to be the point on the ground directly below the point of projection. The ball passes directly over a fielder standing 85 m from the origin.

A wall of height 5 m is located 117.7 m from the origin. The ball is projected with speed $V \text{ m s}^{-1}$ at an angle of elevation θ . Assume that the magnitude of acceleration due to gravity is 10 m s^{-2} . The motion of the ball is modelled by

$$\mathbf{r}(t) = \begin{pmatrix} V \cos \theta t \\ 1 + V \sin \theta t - 5t^2 \end{pmatrix}. \quad (\text{Do NOT prove this.})$$



- (i) On one occasion, Maki hits the ball and it is projected with speed 34.3 m s^{-1} at an angle of elevation of 45° . 3

At the instant the ball is hit, the fielder runs directly towards the wall at a constant speed of $W \text{ m s}^{-1}$.

Assume that the fielder can catch the ball provided its height does not exceed 2 m. Find the minimum value of W required for the fielder to catch the ball before the ball reaches the wall. Give your answer correct to two decimal places.

- (ii) To avoid being caught out, Maki tries to hit the ball over the wall. For a projected speed V and angle of elevation θ , show that the height of the ball when it reaches the wall is given by 1

$$y = 1 + 117.7 \tan \theta - \frac{5(117.7)^2}{V^2} \sec^2 \theta.$$

- (iii) Hence, for a projected speed V , determine the angle of projection which maximises the height of the ball when it reaches the wall. You do **not** need to justify that it is a maximum. 2

————— END OF PAPER —————

SOLUTIONS

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	B	C	B	C	C	D	D	D	C

Question 1 - Answer: C ✓

$$\begin{aligned} |a| &= \sqrt{3^2 + (-2)^2} \\ &= \sqrt{9 + 4} \\ &= \sqrt{13} \end{aligned}$$

Question 2 - Answer: B ✓

The inverse sine function returns principal values only, so its range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

The transformation $x \mapsto x - 1$ changes the domain but does not affect the range.

Therefore, the range of $f(x) = \sin^{-1}(x - 1)$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$.

Question 3 - Answer: C ✓

Two vectors are perpendicular if their dot product is zero.

$$\begin{aligned} \begin{pmatrix} 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \end{pmatrix} &= 2 \times 3 + 3 \times (-2) \\ &= 6 - 6 \\ &= 0 \end{aligned}$$

Therefore the vectors are perpendicular.

Question 4 - Answer: B ✓

By the pigeonhole principle, if each month contained at most 33 pupils with birthdays, there could be at most $12 \times 33 = 396$ pupils.

Therefore, with 397 pupils, at least one month must contain at least 34 pupils with birthdays.

Hence the minimum possible number of pupils is 397.

Question 5 - Answer: C ✓

Given $x = \tan t$ and $y = x^2$, we have $y = \tan^2 t$. Therefore

$$\begin{aligned} \frac{dy}{dt} &= \frac{d}{dt}(\tan^2 t) \\ &= 2 \tan t \sec^2 t \end{aligned}$$

Question 6 - Answer: C ✓

The projection of $\underline{a}$ onto $\underline{b}$ is $\text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|^2} \underline{b}$

Thus

$$\begin{aligned}\lambda &= \frac{\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}}{1^2 + 2^2} \\ &= \frac{4}{5}\end{aligned}$$

Question 7 - Answer: D ✓

The graph of $g(x)$ is obtained by first replacing x with $|x|$, reflecting the right-hand side of the graph of f in the y -axis, and then taking the absolute value, reflecting any negative parts above the x -axis.

Therefore $g(x) = |f(|x|)|$.

Question 8 - Answer: D ✓

For $\frac{dy}{dx} = \frac{y-x}{x}$, horizontal tangents occur where $\frac{y-x}{x} = 0$, that is, along $y = x$.

Also, the differential equation is undefined when $x = 0$, so no line elements should appear on the y -axis.

Furthermore, $\frac{dy}{dx} = -1$ along $y = 0$, and the sign of the gradient changes on either side of the line $y = x$.

The only slope field with these features is D.

Question 9 - Answer: D ✓

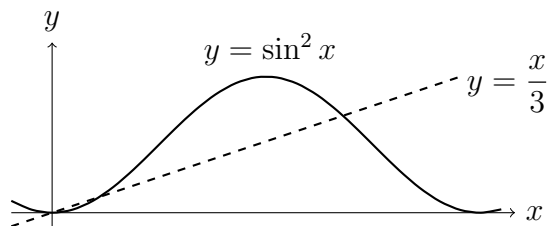
Consider the two youngest members as a single block. Then there are 8 objects to arrange around the circular table.

The number of favourable arrangements is $7! \times 2!$, since the block can be arranged internally in $2!$ ways.

The total number of circular arrangements of 9 people is $(9-1)! = 8!$. Hence the required probability is $\frac{7! \times 2!}{8!}$

Question 10 - Answer: C ✓

Clearly $x = 0$ is a solution.



The graph shows one intersection at $x = 0$, a second for $0 < x < 1$, and a third for $\frac{\pi}{2} < x < 3$.

Since $0 \leq \sin^2 x \leq 1$, there can be no intersections for $x > 3$.

Hence the graph intersects the x -axis three times.

Algebraic approach:

Clearly $x = 0$ is a solution.

Let

$$g(x) = \sin^2 x - \frac{x}{3}$$

Since $0 \leq \sin^2 x \leq 1$, any solution of $g(x) = 0$ must satisfy

$$0 \leq \frac{x}{3} \leq 1$$

so $0 \leq x \leq 3$

Now

$$g(1) = \sin^2(1) - \frac{1}{3} > 0$$

while

$$g\left(\frac{1}{4}\right) = \sin^2\left(\frac{1}{4}\right) - \frac{1}{12} < 0$$

Hence there is a root in $\left(\frac{1}{4}, 1\right)$.

Also,

$$g\left(\frac{\pi}{2}\right) = 1 - \frac{\pi}{6} > 0$$

while

$$g(\pi) = -\frac{\pi}{3} < 0$$

Hence there is another root in $\left(\frac{\pi}{2}, \pi\right)$.

There are no solutions for $x < 0$ and none for $x > 3$.

Therefore the graph intersects the x -axis three times.

Question 11(a)

$${}^7P_4 = \frac{7!}{3!}$$

$$\begin{aligned} {}^7P_4 &= 7 \times 6 \times 5 \times 4 \\ &= \boxed{840} \quad \checkmark \end{aligned}$$

Question 11(b)

$$\begin{aligned} \int_0^1 \frac{2}{1+x^2} dx &= 2 \int_0^1 \frac{1}{1+x^2} dx \\ &= 2 [\tan^{-1} x]_0^1 \quad \checkmark \\ &= 2 \left(\frac{\pi}{4} - 0 \right) \\ &= \boxed{\frac{\pi}{2}} \quad \checkmark \end{aligned}$$

Question 11(c)

Using the substitution $t = \tan \frac{\theta}{2}$, we have $\sin \theta = \frac{2t}{1+t^2}$, and $\cos \theta = \frac{1-t^2}{1+t^2}$.
Hence

$$\begin{aligned} 2 \sin \theta + \cos \theta &= \frac{4t}{1+t^2} + \frac{1-t^2}{1+t^2} \\ &= \frac{1+4t-t^2}{1+t^2} \end{aligned}$$

Therefore

$$\begin{aligned} \frac{1+4t-t^2}{1+t^2} &= 2 \\ 1+4t-t^2 &= 2+2t^2 \\ 3t^2-4t+1 &= 0 \quad \checkmark \\ (3t-1)(t-1) &= 0 \end{aligned}$$

$$t = \frac{1}{3} \quad \text{or} \quad t = 1$$

Since $t = \tan \frac{\theta}{2}$,

$$\tan \frac{\theta}{2} = \frac{1}{3} \quad \text{or} \quad \tan \frac{\theta}{2} = 1$$

$$\frac{\theta}{2} = \tan^{-1} \left(\frac{1}{3} \right) \quad \text{or} \quad \frac{\pi}{4}$$

$\theta = 2 \tan^{-1} \left(\frac{1}{3} \right), \frac{\pi}{2}$	NOTE	$2 \tan^{-1} \left(\frac{1}{3} \right) \approx 0.644 \text{rad}$	✓
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For the second mark, any invalid solutions resulted in the mark not being awarded. Solutions outside of the domain (provided they were correct for a larger domain) were not penalised.

Question 11(d)

$$x = t^2 + 1, \quad y = t^3 - 3t$$

$$\frac{dx}{dt} = 2t,$$

$$\frac{dy}{dt} = 3t^2 - 3$$

Therefore

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

$$= \frac{3t^2 - 3}{2t}$$

When $t = 2$,

$$\frac{dy}{dx} = \frac{3(2)^2 - 3}{2(2)} = \frac{12 - 3}{4} = \frac{9}{4}$$

Hence the gradient of the tangent is

$$\boxed{\frac{9}{4}} \quad \checkmark$$

Question 11(e)

$$A = x^2$$

$$\frac{dA}{dt} = 2x \frac{dx}{dt} \quad \checkmark$$

Given that

$$\frac{dA}{dt} = 12,$$

when $x = 3$,

$$12 = 2(3) \frac{dx}{dt}$$

Hence

$$\frac{dx}{dt} = 2$$

Therefore the side length is increasing at

$$\boxed{2 \text{ cm s}^{-1}} \quad \checkmark$$

Question 11(f)(i)

Using

$$\cos 2x + \sin 2x = R \cos(2x - \alpha),$$

$$R \cos \alpha = 1, \quad R \sin \alpha = 1$$

Hence

$$R = \sqrt{1^2 + 1^2} = \sqrt{2}, \quad \checkmark$$

and

$$\tan \alpha = \frac{1}{1} = 1$$

Since $0 < \alpha < \frac{\pi}{2}$,

$$\alpha = \frac{\pi}{4} \quad \checkmark$$

Therefore

$$\cos 2x + \sin 2x = \sqrt{2} \cos\left(2x - \frac{\pi}{4}\right)$$

Question 11(f)(ii)

$$\cos 2x + \sin 2x = 1$$

$$0 \leq 2x \leq 4\pi$$

$$\sqrt{2} \cos\left(2x - \frac{\pi}{4}\right) = 1$$

$$-\frac{\pi}{4} \leq 2x - \frac{\pi}{4} \leq \frac{15\pi}{4}$$

$$\cos\left(2x - \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}} \quad \checkmark$$

$$2x - \frac{\pi}{4} = -\frac{\pi}{4}, \frac{\pi}{4}, \frac{7\pi}{4}, \frac{9\pi}{4}, \frac{15\pi}{4}$$

$$2x = 0, \frac{\pi}{2}, 2\pi, \frac{5\pi}{2}, 4\pi$$

$$x = 0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}, 2\pi \quad \checkmark \quad \text{all five required}$$

Question 11(g)Let $u = \sqrt{x}$.Then $x = u^2$, $dx = 2u \, du$.When $x = 1$, $u = 1$, and when $x = 4$, $u = 2$.

Therefore

$$\begin{aligned} \int_1^4 \frac{1}{\sqrt{x}(1+\sqrt{x})} dx &= \int_1^2 \frac{1}{u(1+u)} (2u) \, du \quad \checkmark \\ &= 2 \int_1^2 \frac{1}{1+u} \, du \\ &= 2 [\ln(1+u)]_1^2 \quad \checkmark \\ &= 2(\ln 3 - \ln 2) \quad \checkmark \\ &= 2 \ln\left(\frac{3}{2}\right) \end{aligned}$$

$$\ln\left(\frac{9}{4}\right)$$

Question 12(a)

$$\frac{x}{x+1} > 3$$

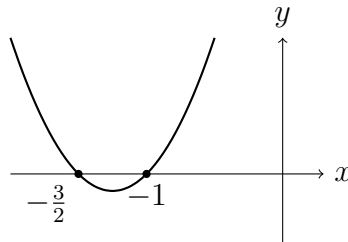
$$x(x+1) > 3(x+1)^2 \quad \checkmark \quad x \neq -1$$

$$0 > 3(x+1)^2 - x(x+1)$$

$$(x+1)(3x+3-x) < 0$$

$$(2x+3)(x+1) < 0.$$

The critical values are $x = -\frac{3}{2}$ and $x = -1$.



Therefore, $-\frac{3}{2} < x < -1 \quad \checkmark \quad \text{OR} \quad (-\frac{3}{2}, -1)$

Question 12(b)(i)

$$\begin{aligned} \underline{a} \cdot \underline{b} &= (-2)(1) + (5)(2) \\ &= 8 \end{aligned}$$

Also,

$$|\underline{a}| = \sqrt{29}, \quad |\underline{b}| = \sqrt{5} \quad \checkmark$$

Hence

$$\begin{aligned} \cos \theta &= \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|} \\ &= \frac{8}{\sqrt{145}} \end{aligned}$$

$$\text{Therefore } \theta = \cos^{-1}\left(\frac{8}{\sqrt{145}}\right) \quad \checkmark$$

Question 12(b)(ii)

Since $\underline{c}$ is perpendicular to $\underline{a}$:

$$\underline{c} \cdot \underline{a} = 0$$

$$(\underline{b} + \lambda \underline{a}) \cdot \underline{a} = 0$$

$$\underline{b} \cdot \underline{a} + \lambda(\underline{a} \cdot \underline{a}) = 0$$

$$8 + 29\lambda = 0$$

$$\lambda = -\frac{8}{29} \quad \checkmark$$

Question 12(b)(iii)

$$\begin{aligned}
\text{Area of parallelogram} &= 2 \times \text{Area of triangle} \\
&= 2 \times \frac{1}{2} |\underline{a}| |\underline{b}| \sin \theta \\
&= |\underline{a}| |\underline{b}| \sin \left(\cos^{-1} \left(\frac{8}{\sqrt{145}} \right) \right) \\
&= \sqrt{145} \times \frac{9}{\sqrt{145}} \\
&= 9 \text{ units}^2 \quad \checkmark
\end{aligned}$$

ALTERNATIVE METHOD:

$$\begin{aligned}
\underline{c} &= \begin{pmatrix} 1 \\ 2 \end{pmatrix} + -\frac{8}{29} \begin{pmatrix} -2 \\ 5 \end{pmatrix} \\
&= \frac{1}{29} \begin{pmatrix} 45 \\ 18 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
|\underline{c}| &= \sqrt{\frac{45^2 + 18^2}{29^2}} \\
&= \frac{9}{\sqrt{29}}
\end{aligned}$$

$$\begin{aligned}
\text{Area of Parallelogram} &= \text{base} \times \text{perpendicular height} \\
&= |\underline{a}| \times |\underline{c}| \\
&= \sqrt{29} \times \frac{9}{\sqrt{29}} \\
&= 9 \text{ units}^2 \quad \checkmark
\end{aligned}$$

Question 12(c)

$$\begin{aligned}
\text{LHS} &= \frac{\sin 2\theta}{1 + \cos 2\theta} \\
&= \frac{2 \sin \theta \cos \theta}{1 + \cos^2 \theta - \sin^2 \theta} \quad \checkmark \\
&= \frac{2 \sin \theta \cos \theta}{2 \cos^2 \theta} \\
&= \frac{\sin \theta}{\cos \theta} \quad \checkmark \\
&= \tan \theta \\
&= \text{RHS}
\end{aligned}$$

ALTERNATIVE METHOD:

Let $t = \tan \theta$, then:

$$\begin{aligned}
 \text{LHS} &= \frac{\sin 2\theta}{1 + \cos 2\theta} \\
 &= \frac{2t}{1 + \frac{1-t^2}{1+t^2}} \quad \checkmark \\
 &= \frac{2t}{1+t^2+1-t^2} \\
 &= \frac{2t}{2} \quad \checkmark \\
 &= t \\
 &= \tan \theta \\
 &= \text{RHS}
 \end{aligned}$$

Question 12(d)

Base Case

For $n = 1$:

$$\begin{aligned}
 7^1 - 1 &= 6 \\
 &= 6 \times 1, \quad \text{hence true for } n = 1 \quad \checkmark
 \end{aligned}$$

Inductive Hypothesis

Assume true for $n = k$:

$$7^k - 1 = 6M \quad \text{where } M \text{ is an integer. } (**)$$

Inductive Step

Prove true for $n = k + 1$:

$$\begin{aligned}
 7^{k+1} - 1 &= 7 \times 7^k - 1 \\
 &= 7(7^k - 1) + 6 \\
 &= 7(6M) + 6 \quad \text{by } (**) \quad \checkmark \\
 &= 6(7M + 1) \quad \checkmark
 \end{aligned}$$

Which is divisible by 6, since $7M + 1$ is an integer.

So if the result is true for $n = k$, then it is true for $n = k + 1$. The result follows by mathematical induction for all positive integers n .

ALTERNATIVE SETTING OUT OF INDUCTIVE STEP:

$$\begin{aligned}
 7^{k+1} - 1 &= 7^k \times 7 - 1 \\
 &= (6M + 1) \times 7 - 1 \quad \text{by } (**) \quad \checkmark \\
 &= 7(6M) + 6 \\
 &= 6(7M + 1) \quad \checkmark
 \end{aligned}$$

Question 12(e)(i)

$$\overrightarrow{AB} = \underline{b} - \underline{a}$$

$$\begin{aligned}\overrightarrow{OP} &= \overrightarrow{OA} + \overrightarrow{AP} \\ &= \underline{a} + \frac{1}{3}(\underline{b} - \underline{a}) \\ &= \frac{2}{3}\underline{a} + \frac{1}{3}\underline{b}\end{aligned}$$

$$\begin{aligned}\overrightarrow{OQ} &= \overrightarrow{OA} + \overrightarrow{AQ} \\ &= \underline{a} + \frac{2}{3}(\underline{b} - \underline{a}) \\ &= \frac{1}{3}\underline{a} + \frac{2}{3}\underline{b}\end{aligned}$$

Thus $\overrightarrow{OP} = \frac{2}{3}\underline{a} + \frac{1}{3}\underline{b}$ and $\overrightarrow{OQ} = \frac{1}{3}\underline{a} + \frac{2}{3}\underline{b}$. ✓

Question 12(e)(ii)

$$\begin{aligned}\overrightarrow{PQ} &= \overrightarrow{OQ} - \overrightarrow{OP} \\ &= \frac{1}{3}\underline{a} + \frac{2}{3}\underline{b} - \left(\frac{2}{3}\underline{a} + \frac{1}{3}\underline{b}\right) \quad \checkmark \text{ 'show that' } \\ &= \frac{1}{3}(\underline{b} - \underline{a})\end{aligned}$$

ALTERNATIVE METHOD:

$$\begin{aligned}\overrightarrow{PQ} &= \overrightarrow{AQ} - \overrightarrow{AP} \\ &= \frac{2}{3}\overrightarrow{AB} - \frac{1}{3}\overrightarrow{AB} \quad \checkmark \text{ 'show that' } \\ &= \frac{1}{3}\overrightarrow{AB} \\ &= \frac{1}{3}(\underline{b} - \underline{a})\end{aligned}$$

Question 12(e)(iii)

$$\begin{aligned}\overrightarrow{OM} &= \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB} \\ &= \underline{a} + \frac{1}{2}(\underline{b} - \underline{a}) \\ &= \frac{1}{2}(\underline{a} + \underline{b}) \quad \checkmark\end{aligned}$$

Now

$$\begin{aligned}\overrightarrow{PM} &= \overrightarrow{OM} - \overrightarrow{OP} \\ &= \frac{1}{2}(\underline{a} + \underline{b}) - \left(\frac{2}{3}\underline{a} + \frac{1}{3}\underline{b}\right) \\ &= \frac{1}{6}(\underline{b} - \underline{a})\end{aligned}$$

Also

$$\begin{aligned}\overrightarrow{MQ} &= \overrightarrow{OQ} - \overrightarrow{OM} \\ &= \frac{1}{3}\underline{a} + \frac{2}{3}\underline{b} - \frac{1}{2}(\underline{a} + \underline{b}) \\ &= \frac{1}{6}(\underline{b} - \underline{a}) \\ &= \overrightarrow{PM}\end{aligned}$$

So $|\overrightarrow{PM}| = |\overrightarrow{MQ}|$ and M is the midpoint of PQ .

ALTERNATIVE METHOD:

$$\begin{aligned}\overrightarrow{PM} &= \overrightarrow{OM} - \overrightarrow{OP} \\ &= \frac{1}{2}(\underline{a} + \underline{b}) - \left(\frac{2}{3}\underline{a} + \frac{1}{3}\underline{b}\right) \\ &= \frac{1}{6}(\underline{b} - \underline{a}) \\ &= \frac{1}{2}\left(\frac{1}{3}(\underline{b} - \underline{a})\right) \\ &= \frac{1}{2}\overrightarrow{PQ}\end{aligned}$$

Question 13(a) The coefficient of x^2 in $(x + a)^8$ is

$$\binom{8}{2}a^6 = 28a^6$$

The coefficient of x^5 is

$$\binom{8}{5}a^3 = 56a^3$$

Equating coefficients,

$$28a^6 = 56a^3 \quad \checkmark$$

$$a^6 = 2a^3$$

$$a^3(a^3 - 2) = 0$$

$a > 0$ Hence

$$\boxed{a = \sqrt[3]{2}} \quad \checkmark$$

Question 13(b)(i) Using the product rule,

$$\frac{d}{dx}(x \tan^{-1}(3x)) = \tan^{-1}(3x) + x \left(\frac{3}{1 + 9x^2} \right)$$

Therefore

$$\boxed{\frac{d}{dx}(x \tan^{-1}(3x)) = \tan^{-1}(3x) + \frac{3x}{1 + 9x^2}} \quad \checkmark$$

Question 13(b)(ii)

From part (i),

$$\tan^{-1}(3x) = \frac{d}{dx}(x \tan^{-1}(3x)) - \frac{3x}{1 + 9x^2}$$

Hence

$$\int \tan^{-1}(3x) dx = x \tan^{-1}(3x) - \int \frac{3x}{1 + 9x^2} dx \quad \checkmark$$

Let

$$u = 1 + 9x^2, \quad du = 18x dx$$

Then

$$\begin{aligned} \int \frac{3x}{1 + 9x^2} dx &= \frac{1}{6} \int \frac{1}{u} du \\ &= \frac{1}{6} \ln u \\ &= \frac{1}{6} \ln(1 + 9x^2) \end{aligned}$$

Therefore

$$\int \tan^{-1}(3x) dx = x \tan^{-1}(3x) - \frac{1}{6} \ln(1 + 9x^2) + C \quad \checkmark$$

13(c)(i)

$$\frac{dP}{dt} = \frac{1}{2}P \left(1 - \frac{P}{2000}\right), \quad P(0) = 300$$

Separate variables:

$$\frac{dP}{P \left(1 - \frac{P}{2000}\right)} = \frac{1}{2} dt$$

$$\text{Since } \frac{1}{P \left(1 - \frac{P}{2000}\right)} = \frac{2000}{P(2000 - P)} = \frac{1}{P} + \frac{1}{2000 - P}$$

$$\int \left(\frac{1}{P} + \frac{1}{2000 - P} \right) dP = \frac{1}{2} \int dt$$

$$\ln |P| - \ln |2000 - P| = \frac{t}{2} + C \quad \checkmark$$

$$\ln \left(\frac{P}{2000 - P} \right) = \frac{t}{2} + C$$

$$\frac{P}{2000 - P} = Ae^{t/2}$$

$$P = Ae^{t/2}(2000 - P)$$

$$P(1 + Ae^{t/2}) = 2000Ae^{t/2}$$

$$P = \frac{2000Ae^{t/2}}{1 + Ae^{t/2}} = \frac{2000}{1 + Be^{-t/2}} \quad \checkmark$$

$$\text{where } B = \frac{1}{A}$$

$$\text{Using } P(0) = 300$$

$$300 = \frac{2000}{1 + B}$$

$$1 + B = \frac{20}{3}$$

$$B = \frac{17}{3}$$

Hence

$$P(t) = \frac{2000}{1 + \frac{17}{3}e^{-t/2}} = \frac{6000}{3 + 17e^{-t/2}} = \frac{6000e^{t/2}}{3e^{t/2} + 17} \quad \checkmark$$

13(c)(ii)

$$\frac{dP}{dt} = \frac{1}{2}P \left(1 - \frac{P}{2000} \right) = \frac{1}{2}P - \frac{P^2}{4000}$$

Treating $\frac{dP}{dt}$ as a quadratic in P , it opens downward, so its maximum occurs at the vertex.

For

$$f(P) = \frac{1}{2}P - \frac{P^2}{4000}$$

the vertex occurs at

$$P = -\frac{b}{2a} = -\frac{\frac{1}{2}}{2\left(-\frac{1}{4000}\right)} = 1000$$

Therefore

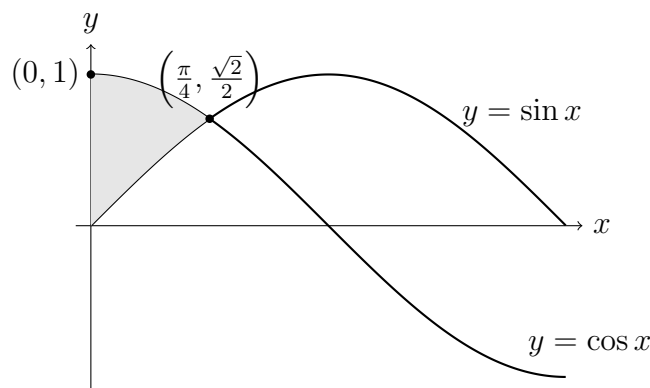
$$\left(\frac{dP}{dt} \right)_{\max} = \frac{1}{2}(1000) \left(1 - \frac{1000}{2000} \right) = 500 \left(\frac{1}{2} \right) = 250$$

Hence

$$\boxed{\left(\frac{dP}{dt} \right)_{\max} = 250} \quad \checkmark$$

Question 13(d)(i)

The curves intersect when $\sin x = \cos x$, so $x = \frac{\pi}{4}$. The required region lies between $y = \cos x$, $y = \sin x$, and the y -axis for $0 \leq x \leq \frac{\pi}{4}$.



Correct shape and intersection. $\checkmark \checkmark$

Correct region indicated. $\checkmark$

Question 13(d)(ii) Rotating about the x -axis gives

$$V = \pi \int_0^{\pi/4} (\cos^2 x - \sin^2 x) dx. \quad \checkmark$$

Using

$$\cos^2 x - \sin^2 x = \cos 2x,$$

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos 2x \, dx && \checkmark \\
 &= \pi \left[\frac{1}{2} \sin 2x \right]_0^{\pi/4} && \checkmark \\
 &= \frac{\pi}{2} (1 - 0).
 \end{aligned}$$

Therefore

$$\boxed{V = \frac{\pi}{2} \text{ units}^3} \quad \checkmark$$

Question 14(a)

The word TATOOINE contains 8 letters, with T repeated twice and O repeated twice.
Total arrangements:

$$\frac{8!}{2!2!} = 10080 \quad \checkmark$$

Arrangements with the two Ts adjacent: Treat TT as a single object. Then there are 7 objects with O repeated twice.

$$\frac{7!}{2!} = 2520$$

Hence the number of arrangements with the Ts separated by one or more letters is

$$10080 - 2520 = 7560$$

$$\boxed{7560} \quad \checkmark$$

Question 14(b)(i)

Since the graph touches, but does not cross, the x -axis at two distinct points, each x -intercept is a repeated root. $\checkmark$

Question 14(b)(ii) Expanding,

$$(x - m)^2(x - n)^2 = x^4 - 2(m + n)x^3 + (m^2 + 4mn + n^2)x^2 - 2mn(m + n)x + m^2n^2.$$

Comparing coefficients with

$$x^4 - 10x^3 + bx^2 + cx + 16,$$

gives

$$2(m + n) = 10 \quad \Rightarrow \quad m + n = 5$$

$$m^2n^2 = 16$$

$$mn = \pm 4$$

Given $mn < 0$,

$$\left. \begin{array}{l} mn = -4 \\ m + n = 5 \end{array} \right\} \quad \text{BOTH} \quad \checkmark$$

Thus m and n satisfy

$$x^2 - 5x - 4 = 0$$

Therefore

$$x = \frac{5 \pm \sqrt{41}}{2} \quad \checkmark$$

Thus

$$\boxed{m = \frac{5 + \sqrt{41}}{2}, \quad n = \frac{5 - \sqrt{41}}{2} \quad \text{OR} \quad m = \frac{5 - \sqrt{41}}{2}, \quad n = \frac{5 + \sqrt{41}}{2}} \quad \checkmark$$

Question 14(c)

$$y = u(x) \cos x$$

$$y' = u' \cos x - u \sin x$$

Substituting into

$$y' \cos x = y \sin x + \cos x \sin x$$

gives

$$(u' \cos x - u \sin x) \cos x = u \cos x \sin x + \cos x \sin x$$

$$u' \cos^2 x - u \sin x \cos x = u \cos x \sin x + \cos x \sin x$$

$$u' \cos^2 x = 2u \cos x \sin x + \cos x \sin x$$

$$u' = 2u \tan x + \tan x$$

✓

$$u' = (2u + 1) \tan x$$

$$\frac{du}{2u + 1} = \tan x \, dx$$

$$\int \frac{du}{2u + 1} = \int \tan x \, dx$$

$$\frac{1}{2} \ln(2u + 1) = -\ln(\cos x) + C$$

$$\ln(2u + 1) = -2 \ln(\cos x) + C$$

$$2u + 1 = A \sec^2 x$$

$$u = \frac{A \sec^2 x - 1}{2}$$

✓

Since $y = u \cos x$

$$y = \frac{1}{2} (A \sec^2 x - 1) \cos x = \frac{1}{2} (A \sec x - \cos x)$$

$y = \frac{A \sec x - \cos x}{2}$

✓

Question 14(d)(i)

With $V = 34.3$ and $\theta = 45^\circ$,

$$x(t) = Vt \cos \theta, \quad y(t) = 1 + Vt \sin \theta - 5t^2$$

The ball is catchable while $y \leq 2$.

Setting $y = 2$,

$$5t^2 - Vt \cos \theta + 1 = 0$$

The later solution is

$$t \approx 4.80198 \text{ s} \quad \checkmark$$

Let W be the minimum constant speed of the fielder. Since the fielder starts at $x = 85$ m, their position after t seconds is

$$85 + Wt$$

To make the catch,

$$85 + Wt = Vt \cos \theta$$

so

$$Wt = Vt \cos \theta - 85 = \frac{34.3}{\sqrt{2}}t - 85 \quad \checkmark$$

Using $t = 4.80198$,

$$W = \frac{31.64195}{4.80198} \approx 6.59$$

$$\boxed{W_{\min} \approx 6.59 \text{ m s}^{-1}} \quad \checkmark$$

(Accept answers that round to 6.6 m s^{-1} .)

Question 14(d)(ii)

At the wall,

$$117.7 = Vt \cos \theta$$

$$t = \frac{117.7}{V \cos \theta}$$

Substituting into

$$y = 1 + Vt \sin \theta - 5t^2$$

gives

$$y = 1 + 117.7 \tan \theta - \frac{5(117.7)^2}{V^2 \cos^2 \theta} \quad \checkmark$$

Therefore

$$\boxed{y = 1 + 117.7 \tan \theta - \frac{5(117.7)^2}{V^2} \sec^2 \theta}$$

Question 14(d)(iii)

Using

$$\sec^2 \theta = 1 + \tan^2 \theta$$

let

$$u = \tan \theta$$

Then

$$y = 1 + 117.7u - \frac{5(117.7)^2}{V^2}(1 + u^2)$$

Differentiating,

$$\frac{dy}{du} = 117.7 - \frac{10(117.7)^2}{V^2}u \quad \checkmark$$

Setting $\frac{dy}{du} = 0$

$$u = \frac{V^2}{10(117.7)}$$

Since $u = \tan \theta$,

$$\boxed{\theta = \tan^{-1}\left(\frac{V^2}{1177}\right)} \quad \checkmark$$